

UNICAST ROUTING

- If a datagram is destined for only one destination (***one-to-one delivery***) , we have unicast routing.
- Routing Tables.
- A packet is routed, hop by hop, from its source to destination by the help of forwarding tables.

UNICAST ROUTING

- To find the best route, an internet can be modelled as a graph.
- A set of nodes and edges that connect the nodes.
- Router – As a node
- Each network between two nodes – As an edge.

UNICAST ROUTING

Least cost routing

Total cost for the route is the least cost among all possible routes.

This means that, each router needs to find the least cost route between itself and all the other routers to be able to route a packet using this criteria.

UNICAST ROUTING

Least-Cost Trees

There are N routers in an internet.

There are $(N - 1)$ least cost paths from each router to any other router.

Means we need $N \times (N - 1)$ least cost paths for the whole internet.

Better way to find the least cost path is, ***Least-Cost Trees***.

UNICAST ROUTING

Least-Cost Trees

A tree with source router as the root that spans the whole graph (visits all other nodes) and in which the path between the route and any other node is the shortest.

ROUTING ALGORITHMS

Distance Vector Routing

A distance vector routing algorithm operates by having each router maintain a table (i.e., a vector) giving the best known distance to each destination and which link to use to get there.

These tables are updated by exchanging information with the neighbours.

Eventually, every router knows the best link to reach each destination.

ROUTING ALGORITHMS

Bellman-Ford Equation

Used to find the least cost (***shortest distance***) ***between a source node x and a destination node y***, through some intermediary nodes (a, b, c, etc.).

$$D_{xy} = \min \{ (c_{xa} + D_{ay}), (c_{xb} + D_{by}), (c_{xc} + D_{cy}), \dots \}$$

Where D_{ij} is the shortest distance;

C_{ij} is the cost between the nodes i and j

ROUTING ALGORITHMS

Bellman-Ford Equation

In distance vector routing, each router maintains a routing table indexed by, and containing one entry for each router in the network.

This entry has two parts:

1. The preferred outgoing line to use for that destination.
 2. An estimate of the distance to that destination.
- The distance might be measured as the number of hops or using another metric, as we discussed for computing shortest paths.

ROUTING ALGORITHMS

- The router is assumed to know the “distance” to each of its neighbours.
- If the metric is hops, the distance is just one hop.
- If the metric is propagation delay, the router can measure it directly with special ECHO packets that the receiver just timestamps and sends back as fast as it can.

ROUTING ALGORITHMS

Bellman-Ford Equation

Information kept by DV router

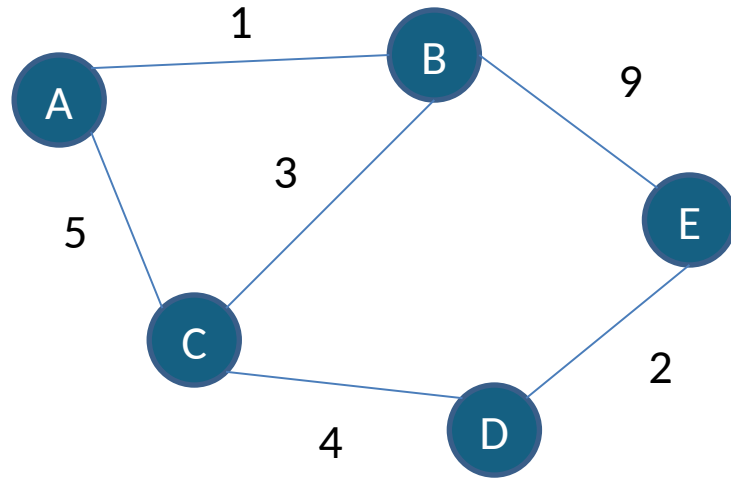
- Each router has an ID
- Associated with each link connected to a router, there is a link cost (static or dynamic).
- Intermediate hops

Distance Vector Table Initialization

- Distance to itself = 0
- Distance to ALL other routers = infinity number.

ROUTING ALGORITHMS

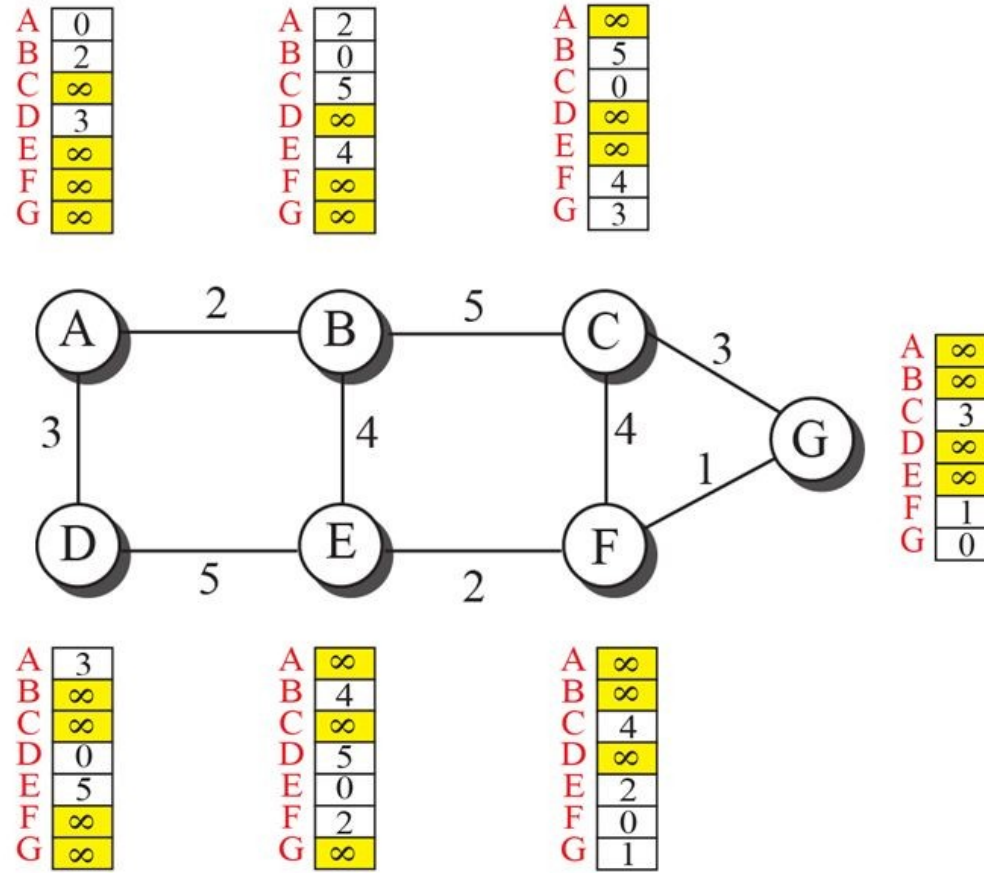
Routing Table



Destin ation	Cost	Next Hop
A	1	A
C	3	C
E	9	E

Initial Routing Table of B

ROUTING ALGORITHMS



ROUTING ALGORITHMS

New B		Old B		A	
A	2	A	2	A	0
B	0	B	0	B	2
C	5	C	5	C	∞
D	5	D	∞	D	3
E	4	E	4	E	∞
F	∞	F	∞	F	∞
G	∞	G	∞	G	∞

$B[j] = \min(B[j], 2 + A[j])$

a. First event: B receives a copy of A's vector.

Note:

$X[]$: the whole vector

New B		Old B		E	
A	2	A	2	A	∞
B	0	B	0	B	4
C	5	C	5	C	∞
D	5	D	5	D	5
E	4	E	4	E	0
F	6	F	∞	F	2
G	∞	G	∞	G	∞

$B[j] = \min(B[j], 4 + E[j])$

b. Second event: B receives a copy of E's vector.

LINK-STATE ROUTING

Distance vector routing was used in the ARPANET until 1979, when it was replaced by link state routing.

- The primary problem that caused its demise was that the algorithm often took too long to converge after the network topology changed.
- Variants of link state routing called IS-IS and OSPF are the routing algorithms that are most widely used inside large networks and the Internet today.

LINK-STATE ROUTING

The idea behind link state routing is fairly simple and can be stated as five parts.

Each router must do the following things to make it work:

1. Discover its neighbors and learn their network addresses.
2. Set the distance or cost metric to each of its neighbors.
3. Construct a packet telling all it has just learned.
4. Send this packet to and receive packets from all other routers.
5. Compute the shortest path to every other router.

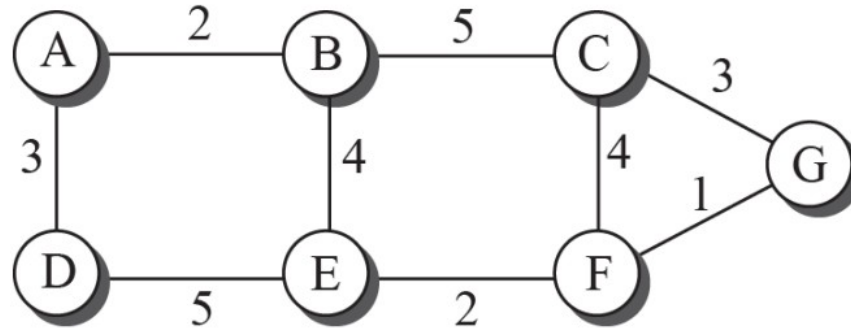
LINK-STATE ROUTING

In effect, the complete topology is distributed to every router.

Then Dijkstra's algorithm can be run at each router to find the shortest path to every other router.

LINK-STATE ROUTING

LSDB (Least Cost DataBase)



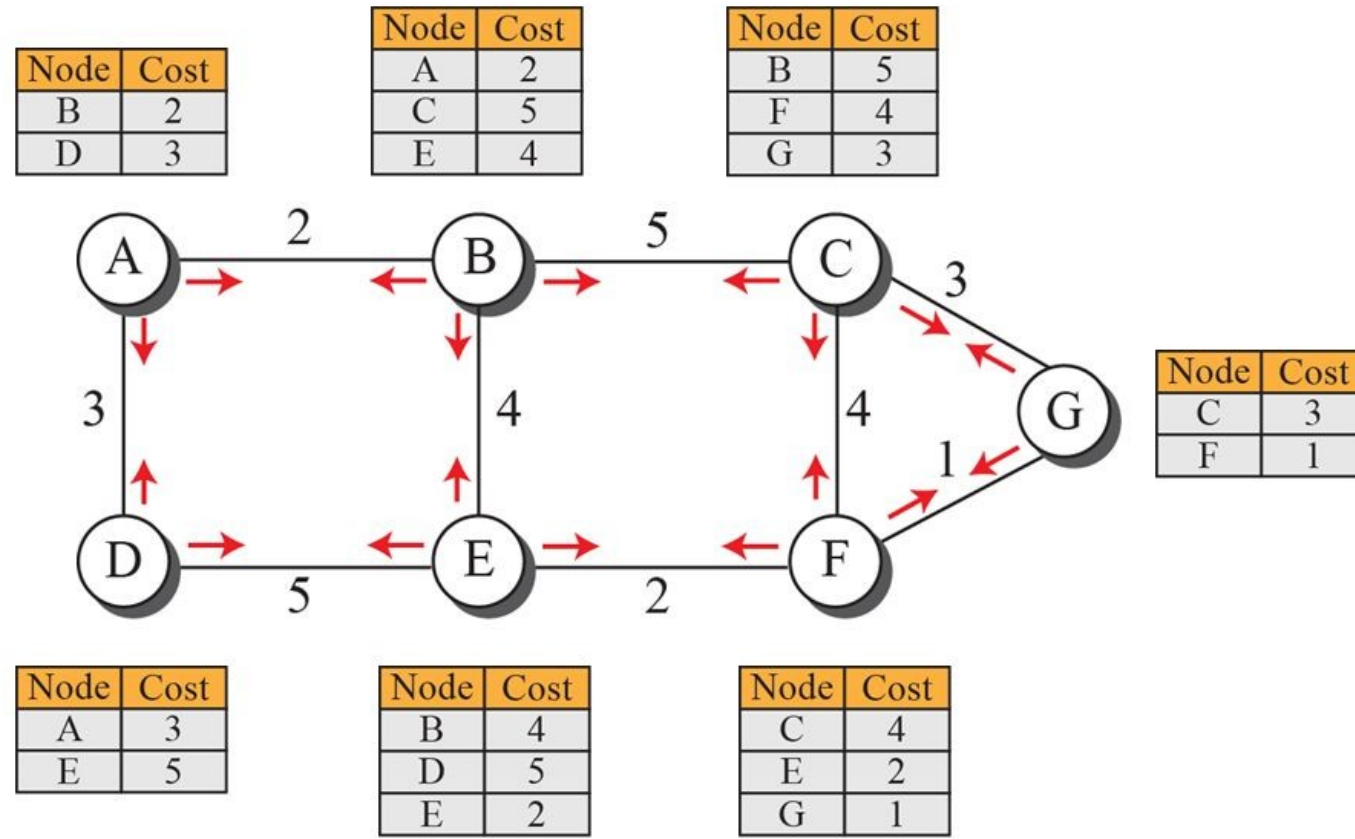
a. The weighted graph

	A	B	C	D	E	F	G
A	0	2	∞	3	∞	∞	∞
B	2	0	5	∞	4	∞	∞
C	∞	5	0	∞	∞	4	3
D	3	∞	∞	0	5	∞	∞
E	∞	4	∞	5	0	2	∞
F	∞	∞	4	∞	2	0	1
G	∞	∞	3	∞	∞	1	0

b. Link state database

LINK-STATE ROUTING

LSDB (Least Cost DataBase)



LINK-STATE ROUTING

Least-Cost Trees

Created using Dijkstra's Algorithm.

Steps:

1. The node chooses itself as the root node of the tree.
2. The node selects one node, among all nodes in the tree, which is closest to the root, and adds this to the tree. After this node is added to the tree, the cost of all other nodes not in the tree needs to be updated because the paths may have been changed.
3. The node repeats step 2 until all nodes are added to the tree.

PATH VECTOR ROUTING

Both previous protocols were based on least cost method.

To overcome the disadvantages of both LS and DV, Path Vector routing came in to existence.

Here, the best route is selected by the source using the policy it imposes on the route.

That means, the source can control the path.

PATH VECTOR ROUTING

Spanning Trees

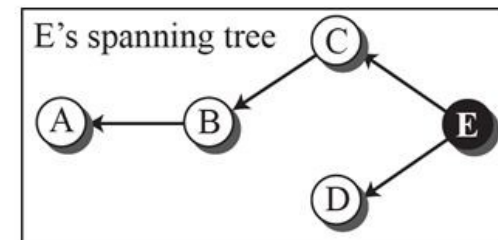
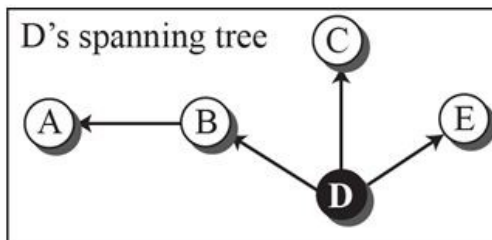
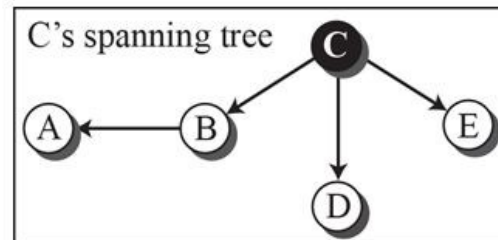
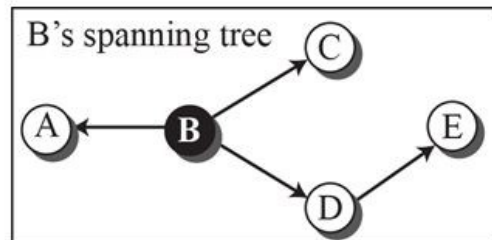
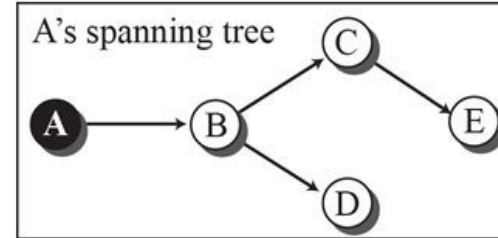
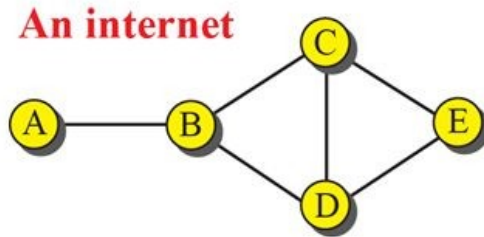
The path is determined by the best spanning tree.

It is not the least cost tree.

It is the tree determined by the source when it imposes its own policy.

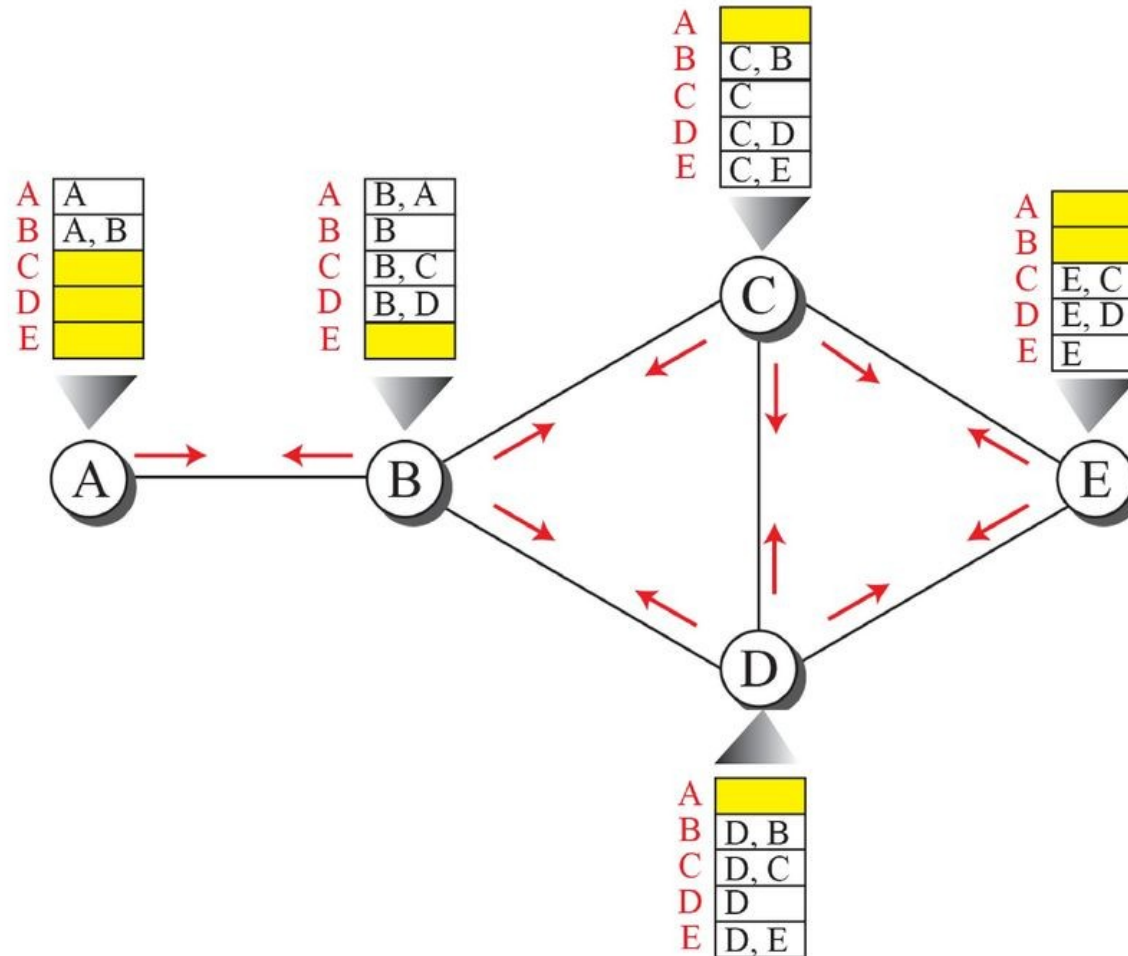
PATH VECTOR ROUTING

Spanning Trees



PATH VECTOR ROUTING

Spanning Trees



PATH VECTOR ROUTING

	New C	Old C	B
A	C, B, A		B, A
B	C, B	C, B	B
C	C	C	B, C
D	C, D	C, D	B, D
E	C, E	C, E	

$C[] = \text{best}(C[], C + B[])$

Note:
 $X[]$: vector X
 Y: node Y

Event 1: C receives a copy of B's vector

	New C	Old C	D
A	C, B, A	C, B, A	
B	C, B	C, B	D, B
C	C	C	D, C
D	C, D	C, D	D
E	C, E	C, E	D, E

$C[] = \text{best}(C[], C + D[])$

Event 2: C receives a copy of D's vector

IPV4 ADDRESSES

- **IP Address:** It is an identifier, which is used to identify the connection of each device to the internet.
- It's a 32 bit address that uniquely and universally defines the connection of a host or a router to the internet.
- It is the address of the connection, not the host or the router.
- If the device is moved to another network, the IP address may be changed.

IPV4 ADDRESSES

- **IPV4 address** are unique in the sense, that each address defines one and only one, connection to the internet.
- If a device has two connections to the internet, via two networks, it has two IPV4 addresses.
- IPV4 addresses are universal in the sense that the addressing system must be accepted by any host that wants to be connected to the internet.

IPV4 ADDRESSES

Address Space

A protocol like IPV4 that defines addresses has an address space.

An address space is the total number of addresses used by the protocol.

If a protocol uses b bits to define an address, the address space is 2^b because each bit can have two different values (0 or 1).

IPV4 uses 32 bit addresses, which means that the address space is 2^{32} **or 4,294,967,296** (more than four billion).

IPV4 ADDRESSES

Notation

There are three common notations to show an IPV4 address:

- Binary notation(base 2)
- Dotted-decimal notation (base 256)
- Hexadecimal notation (base 16)

IPV4 ADDRESSES

Notation

In binary notation – it is displayed as 32 bits.

Each octet is referred to as a byte.

To make the IPV4 more compact and easier to read, it is usually written in decimal form with a decimal point separating the bytes.

This format is known as dotted-decimal notation.

IPV4 ADDRESSES

Notation

Note that because each byte is only 8 bits (octet), each number in the dotted-decimal notation is between 0 and 255.

We sometimes see an IPV4 in hexadecimal notation.

Each hexadecimal digit is having 4 bits.

This means that a 32 bit address has 8 hexadecimal digits.

This notation is often used in network programming.

IPV4 ADDRESSES

Notation

Note that because each byte is only 8 bits (octet), each number in the dotted-decimal notation is between 0 and 255.

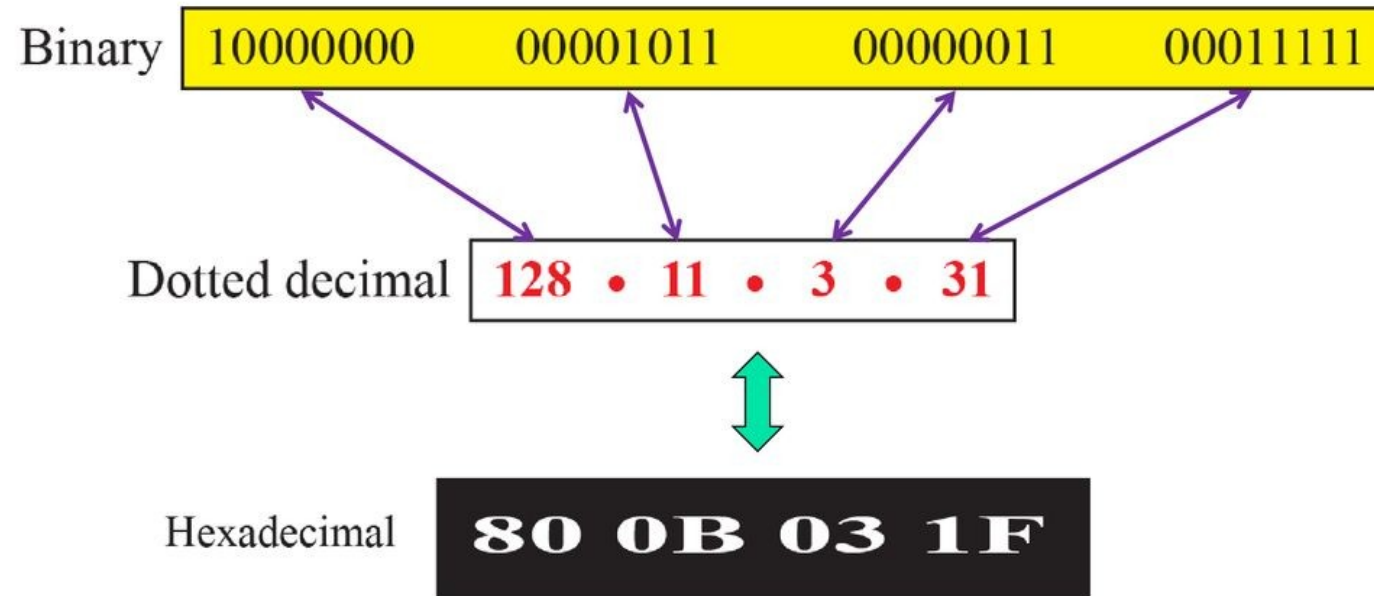
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IPV4 ADDRESSES



IPV4 ADDRESSES

Hierarchy in addressing

The addressing system is hierarchical.

A 32 bit IPV4 address is divided into two parts:

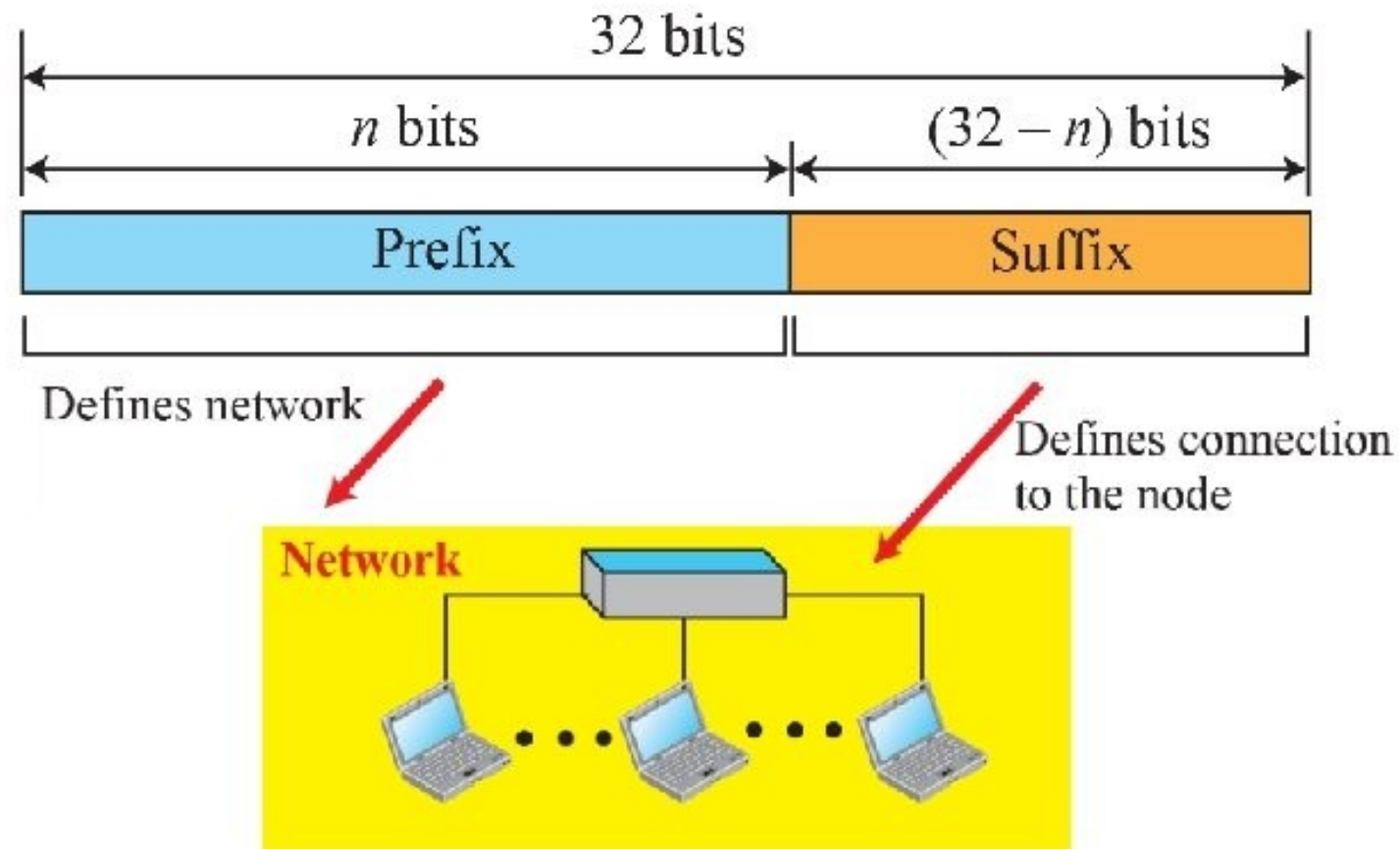
Prefix and suffix

Prefix defines the network and suffix defines the node.

The network identifier in the IPV4 was first designed as a fixed-length prefix, and is referred to as classful addressing.

The new scheme, which is referred to as classless addressing, uses a variable length network prefix.

IPV4 ADDRESSES



IPV4 ADDRESSES

CLASSFUL ADDRESSING

When internet started, the IPV4 address was designed with a fixed length prefix.

But to accommodate both small and large networks, three fixed-length prefixes were designed instead of one. ($n = 8$, $n = 16$, and $n = 24$).

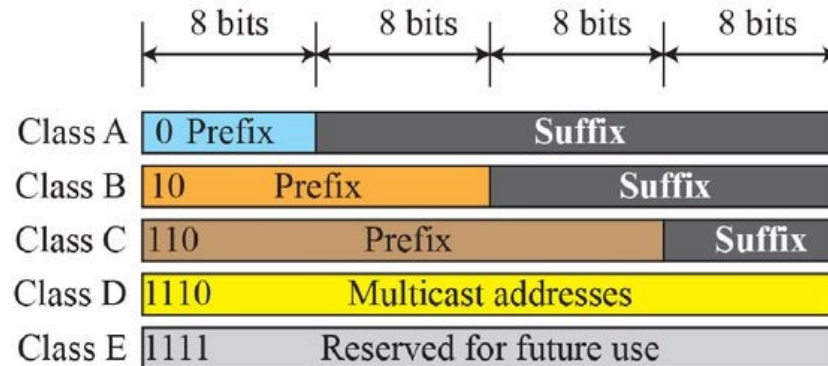
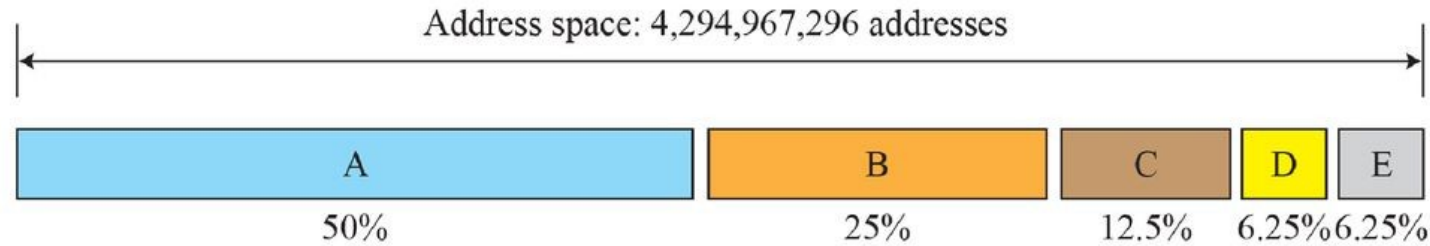
The whole address space was divided into five classes.

CLASS A, CLASS B, CLASS C, CLASS D and CLASS E

This scheme is referred to as classful addressing.

IPV4 ADDRESSES

CLASSFUL ADDRESSING



Class	Prefixes	First byte
A	$n = 8$ bits	0 to 127
B	$n = 16$ bits	128 to 191
C	$n = 24$ bits	192 to 223
D	Not applicable	224 to 239
E	Not applicable	240 to 255

IPV4 ADDRESSES

CLASSLESS ADDRESSING

Subnetting and supernetting in classful addressing did not really solve the address depletion problem.

With the growth of the internet, it was clear that a larger address space was needed as a long term solution.

The larger address space, however, requires that the length of IP addresses also be increased.

It means that the format of the Ip Packets needs to be changed.

IPV4 ADDRESSES

CLASSLESS ADDRESSING

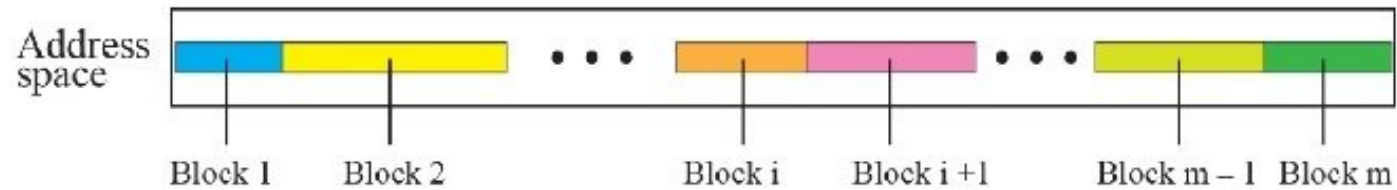
The short term solution for solving this problem was called as classless addressing.

In other words, the class privilege was removed from the distribution to compensate for the address depletion.

In classless addressing variable length blocks are used that belong to no classes.

IPV4 ADDRESSES

CLASSLESS ADDRESSING



INTERNET PROTOCOL (IP)

The network layer in version 4 can be thought of as one main protocol and three auxiliary ones.

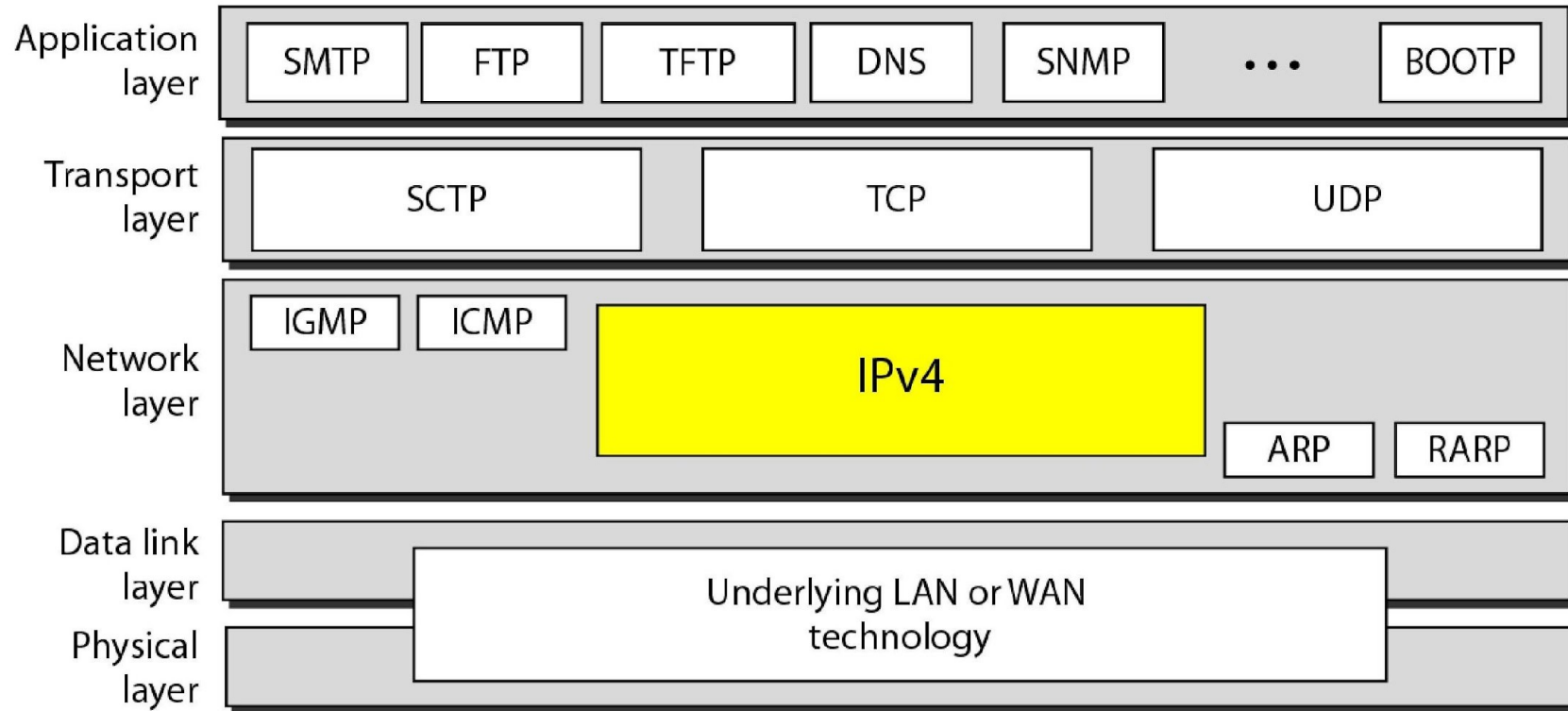
the main protocol, internet protocol version 4 (IPv4), is responsible for packetizing, forwarding and delivery of a packet at the network layer.

The ICMPv4 helps IPv4 to handle some errors that may occur in the network layer delivery.

The IGMP is used to help IPv4 in multicasting.

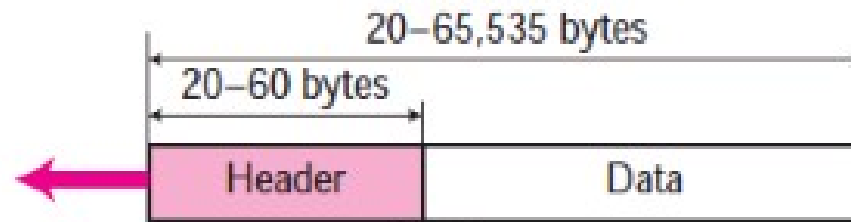
The ARP is used to glue the network and data-link layers in mapping network layer addresses to link-layer addresses.

INTERNET PROTOCOL (IP)

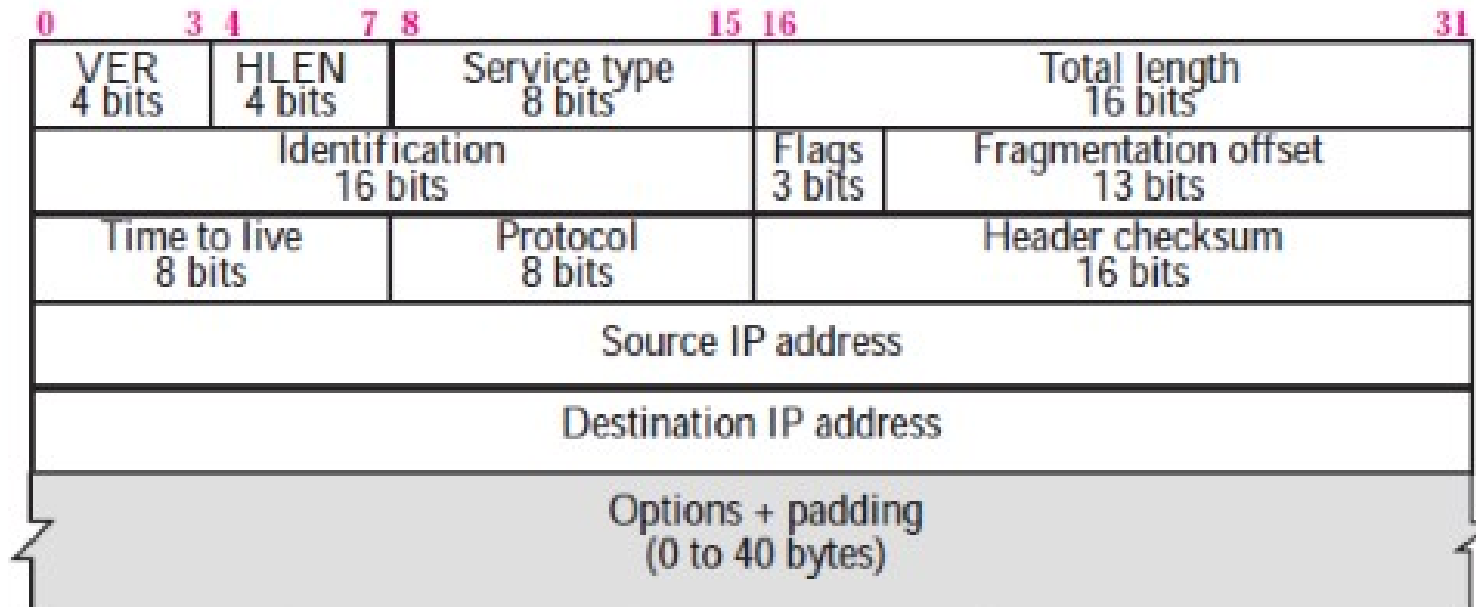


INTERNET PROTOCOL (IP)

DATAGRAM FORMAT



a. IP datagram



b. Header format